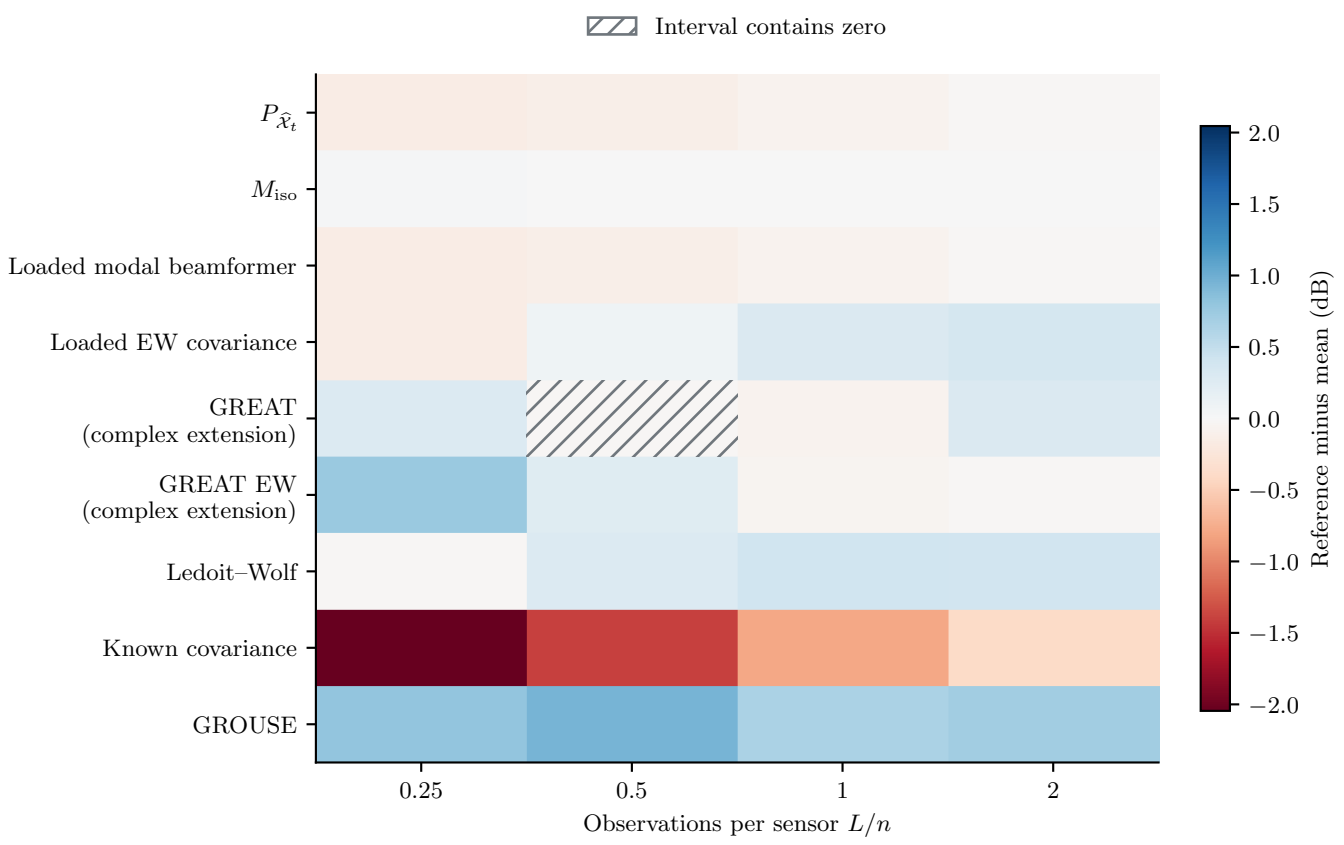
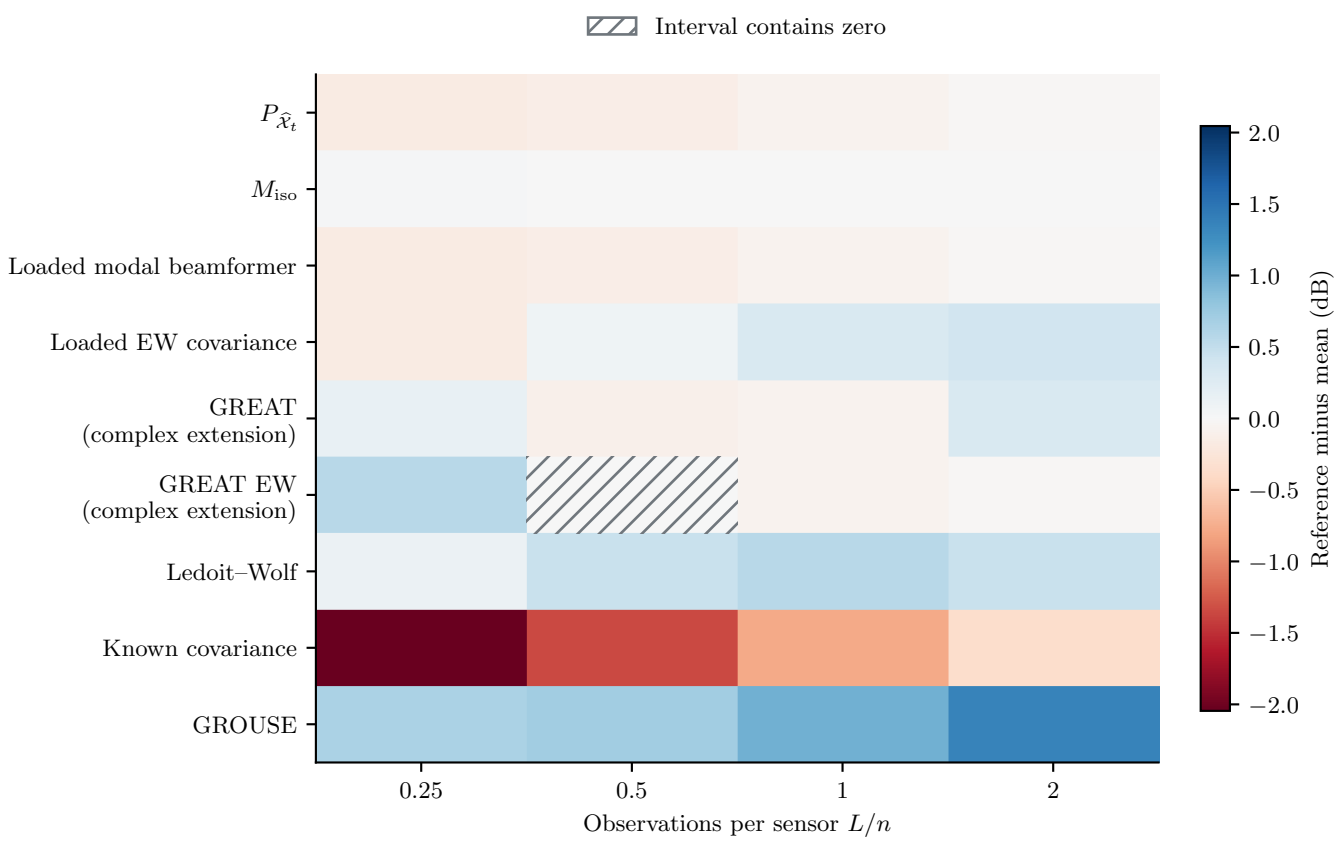
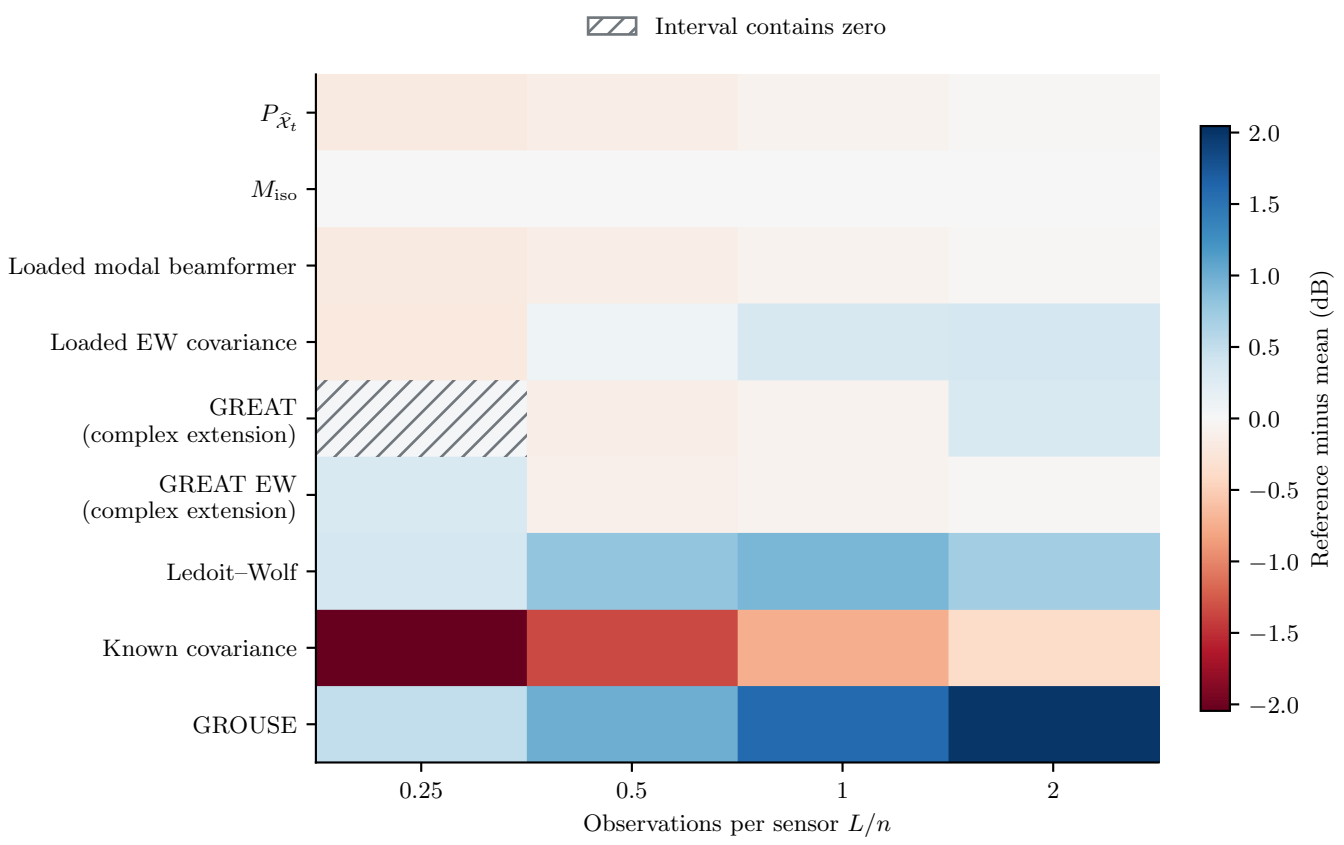
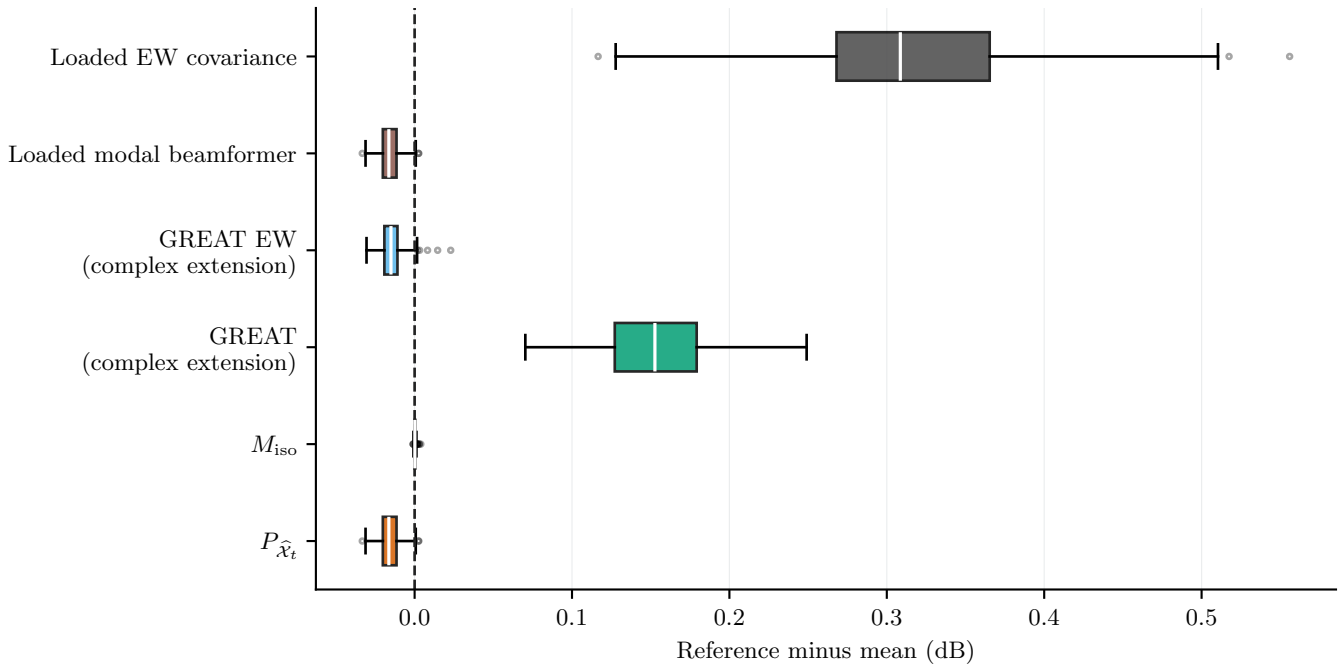
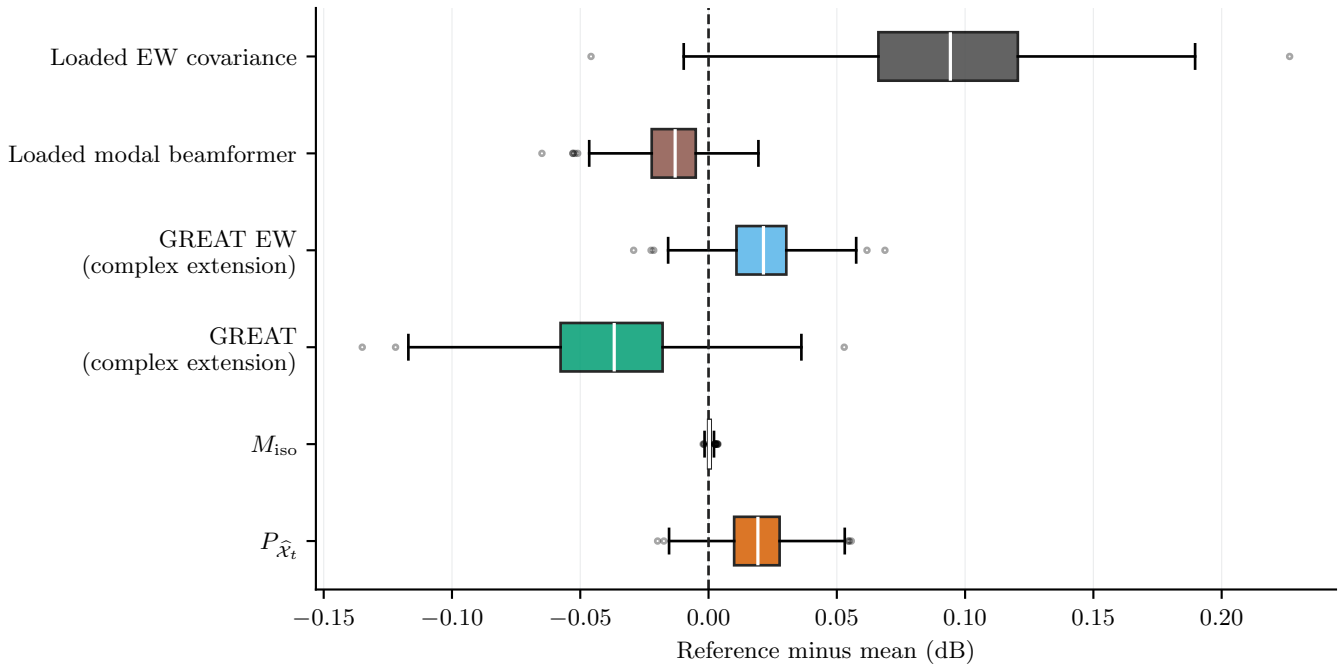
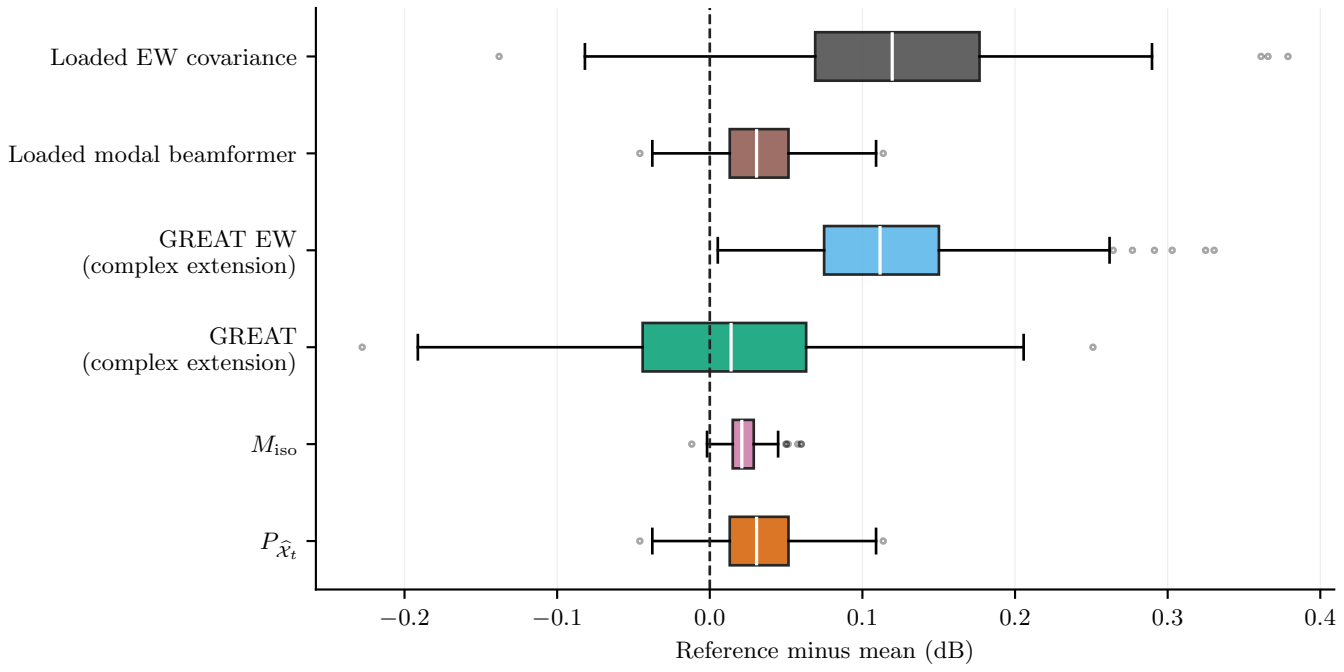

Interval contains zero
$P_{\widehat{\mathcal{X}}_t}$
$M_{\text{iso}}$
Loaded modal beamformer
Loaded EW covariance
GREAT (complex extension)
GREAT EW (complex extension)
Ledoit–Wolf
Known covariance
GROUSE
0.25
0.5
1
2
Observations per sensor $L/n$
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Reference minus mean (dB)

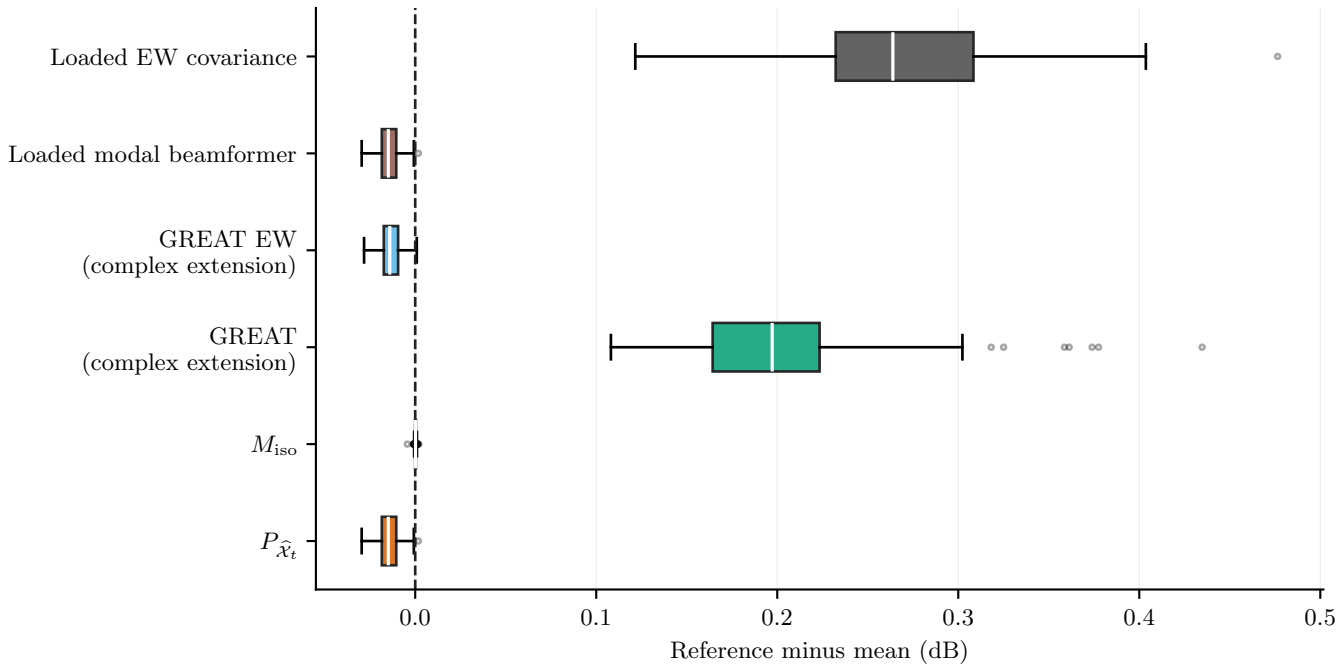

Interval contains zero
$P_{\widehat{\mathcal{X}}_t}$
$M_{\mathrm{iso}}$
Loaded modal beamformer
Loaded EW covariance
GREAT (complex extension)
GREAT EW (complex extension)
Ledoit–Wolf
Known covariance
GROUSE
0.25
0.5
1
2
Observations per sensor $L/n$
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Reference minus mean (dB)

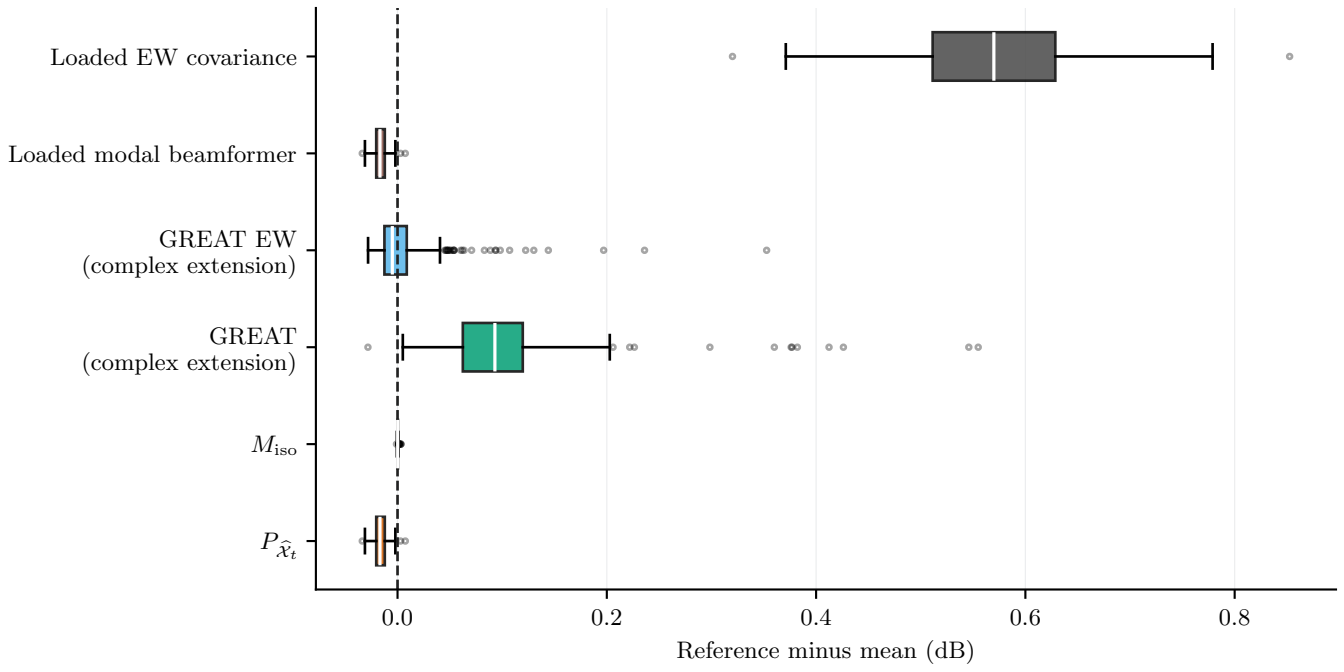

Interval contains zero
$P_{\widehat{\mathcal{X}}_t}$
$M_{\text{iso}}$
Loaded modal beamformer
Loaded EW covariance
GREAT
(complex extension)
GREAT EW
(complex extension)
Ledoit–Wolf
Known covariance
GROUSE
0.25
0.5
1
2
Observations per sensor $L/n$
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Reference minus mean (dB)


Loaded EW covariance
Loaded modal beamformer
GREAT EW
(complex extension)
GREAT
(complex extension)
$M_{\mathrm{iso}}$
$P_{\widehat{\mathcal{X}}_t}$
0.0
0.1
0.2
0.3
0.4
0.5
Reference minus mean (dB)


Loaded EW covariance
Loaded modal beamformer
GREAT EW
(complex extension)
GREAT
(complex extension)
$M_{\text{iso}}$
$P_{\widehat{\mathcal{X}}_t}$
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
Reference minus mean (dB)


Loaded EW covariance
Loaded modal beamformer
GREAT EW
(complex extension)
GREAT
(complex extension)
$M_{\mathrm{iso}}$
$P_{\widehat{\mathcal{X}}_t}$
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
Reference minus mean (dB)


Loaded EW covariance
Loaded modal beamformer
GREAT EW
(complex extension)
GREAT
(complex extension)
$M_{\mathrm{iso}}$
$P_{\widehat{\mathcal{X}}_t}$
0.0
0.1
0.2
0.3
0.4
0.5
Reference minus mean (dB)


Loaded EW covariance
Loaded modal beamformer
GREAT EW
(complex extension)
GREAT
(complex extension)
$M_{\mathrm{iso}}$
$P_{\widehat{\mathcal{X}}_t}$
0.0
0.2
0.4
0.6
0.8
Reference minus mean (dB)